Course Code: CS2201 **Name of the Programme: B.Sc. Computer Science (Hons.)** **Name of the Course: Database Management Systems** **Semester: II**		
Course credit 3	**No. of hours per week (L+ T+P)**	**Total no. of Teaching hours:**
3	**1+1+2**	**60**

Course Objectives:	
a)	Understand the Core Concepts of Database Management Systems
b)	Apply Data Modeling Techniques to Design Efficient Databases
c)	Develop Proficiency in SQL for Data Manipulation and Querying
d)	Demonstrate practical skills in implementing and managing databases while analyzing their role in real-world applications.

Course outline (Syllabus of the course)

Syllabus:

Module - 1	**No. of hours**
	6
Introduction to Database Systems: Definition and Importance of Databases, Database system architecture, Characteristics of DBMS vs. File Systems, Advantages & Disadvantages of DBMS, Types of Databases (Hierarchical, Network, Relational), Data Independence, Database Users and Administrators, Three-tier Architecture (External, Conceptual, Internal Levels) Data Models: Hierarchical, Network, Relational, Object-Oriented.	

Module - 2	**No. of hours**
	6
Data modelling using ER-Model and Constraints: Entity Types, Entity Sets, Attributes, and Keys, Relationship Types, Introduction to ER Diagrams, Components of ER Diagrams, Drawing ER Diagrams, Specialization, Generalization & Aggregation, Mapping ER Diagrams to Relational Schema, Integrity constraints, entity integrity, referential integrity, Keys constraints, Domain constraints.	

Module - 3	**No. of hours**
	9
SQL Queries: Database language and interfaces - DDL, DML, DCL and TCL, Characteristics of SQL, advantages of SQL. SQL data type and literals, Types of SQL commands, SQL operators and their procedure, Tables, views and indexes, Queries and subqueries, Aggregate functions. Insert, update and delete operations, Joins, Unions, Intersection, Minus, Triggers and Procedures in SQL.	

Module - 4	**No. of hours**
	4
Normalization: Introduction to Normalization, Data Anomalies, Functional Dependencies, Types of Normal Forms, 1Normal Form, 2 Normal Form, 3 Normal Form, BCNF(Boyce & Codd Normal Form.).	

Module - 5	**No. of hours**
	5
NOSQL: Introduction to NoSQL, Types of NoSQL Databases, Advantages & Disadvantages of NoSQL, Basics of MongoDB, Querying in NoSQL (MongoDB)	

Course outcomes
a) Explore core database concepts, including architecture and transactions in DBMS.
b) Design and implement database applications using ER diagrams to solve real-world problems.
c) Apply query processing techniques and normalization to optimize database performance and eliminate redundancy.
d) Understand the fundamentals and applications of NoSQL databases for handling non-relational data.

Text Books	
1	Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database system concepts* (7th ed.). McGraw-Hill. ISBN 978-9390727506
2	Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson. ISBN 978-9332582705
3	Sadalage, P. J., & Fowler, M. (2012). *NoSQL distilled: A brief guide to the emerging world of polyglot persistence*. Addison-Wesley Professional. ISBN 978-0321826626

Reference Books	
1	Ramakrishnan, R., & Gehrke, J. (2002). *Database management systems*. McGraw-Hill Higher Education. ISBN 0072465638
2	Vaswani, V. (2004). *MySQL: The complete reference*. Osborne/McGraw-Hill. ISBN 0072224770
3	Sullivan, D. (2015). *NoSQL for mere mortals*. Addison-Wesley. ISBN 0134023218
4	Hows, D., Membrey, P., & Plugge, E. (2014). *MongoDB basics*. Apress. ISBN 148420896X

Tutorials	**[15 Hrs]**
1	Activity to design a University Management System using DDL commands (CREATE, ALTER, and DROP) in MySQL. The database will consist of 5 tables and cover different DDL operations. Ensure the usage of Primary Keys and Foreign Keys. **Modifying Tables** Add a new column PhoneNumber (VARCHAR, 15) to the Students table. Modify the CourseName column in Courses to allow up to 150 characters. Remove the Salary column from the Professors table.
2	Perform DML (Data Manipulation Language) operations on a database designed for an online shopping system. This activity covers INSERT, UPDATE, DELETE, and SELECT statements. Create a database named OnlineShoppingDB, Create the following tables: Customers, Products, Stores, Orders: Stores order details, OrderDetails, Payments: Insert 5 records for all the tables, Update the stock quantity of a product after a sale. Increase the price of all products in the "Furniture" category by 10%. Retrieve all the records from the database.

3	This activity will help students understand and apply SQL functions (String, Numeric, Date, and Aggregate functions) using a simple School Database scenario.A school database stores information about students and their test scores. You will use various SQL functions to analyze and retrieve meaningful insights from the data.
4	Activity: Implementing Date and Conversion Functions in a School Database This activity focuses on using SQL Date and Conversion functions in a School Database scenario to manipulate and retrieve meaningful date-related information. A school database stores information about students and their test scores. You will use Date and Conversion functions to extract and analyze data based on students' birth dates, test dates, and numerical/textual conversions.
5	Activity: Implementing Arithmetic, Logical, and Comparison Operators in SQL This activity helps students understand and apply arithmetic, logical, and comparison operators in SQL using a Retail Store Database scenario. A retail store manages customer orders and tracks product inventory. The store wants to analyze data using arithmetic operations (calculations), logical conditions (AND, OR, NOT), and comparisons (>, <, =, etc.).
6	A retail store wants to analyze its product inventory and customer orders using special operators and set operations. The store needs to: ● Find specific products based on patterns, price ranges, and availability. ● Retrieve unique customer purchases using set operations.
7	Activity to implement the different types of join for the given scenario Scenario: "Online Bookstore Management" A bookstore manages books, customers, and orders in a database. The store wants to generate reports using JOINs to: ● Find which customers ordered which books. ● List books along with their categories.

	• Retrieve order details with customer names.
8	Activity for Implementing GROUP BY, HAVING, and ORDER BY in MySQL *"Restaurant Billing System"* A restaurant keeps track of customer orders, including food items, prices, and quantities. The manager wants to: • Find total sales per item. • Filter items based on total revenue. • Sort the results by revenue.
9	Activity: Implementing a Stored Procedure in MySQL *Bank Account Management* A bank wants to automate common operations on customer accounts, such as adding new customers, updating balances, and deleting customer records when needed.Create a stored procedure AddCustomer to insert a new customer into the Customers table.Create a stored procedure UpdateBalance to update an account balance.Create a stored procedure DeleteCustomer to remove a customer from the database
10	Activity Implementing Triggers in MySQL for INSERT, UPDATE, and DELETE *Bank Account Management* A bank wants to keep track of changes to customer accounts automatically whenever new accounts are added, balances are updated, or accounts are deleted. Implement triggers to handle these events. Create a trigger AfterInsert_Customer, AfterUpdate_Balance , BeforeDelete_Customer

Lab Programs	**[30 Hrs]**
1	Implementation of DDL Commands in SQL: Demonstrate the use of DDL commands with suitable examples, including CREATE TABLE, ALTER TABLE, and DROP TABLE.

2	Implementation of DML Commands in SQL: Implement DML commands with suitable examples, including INSERT, UPDATE, and DELETE.
3	Implementation of SQL Functions: Implement various types of functions in SQL with examples, including Number Functions, Aggregate Functions, Character Functions.
4	Implementation of SQL Functions: Conversion Functions, and Date Functions.
5	Implementation of SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.
6	Implementation of SQL Operators: Demonstrate the use of different types of operators in SQL with examples Special Operators, and Set Operations(UNION)
7	Implementation of Different Types of Joins: Implement various types of joins in SQL, including Inner Join, Outer Join, and Natural Join, with suitable examples.
8	Study and Implementation of Grouping and Ordering: Study and implement the GROUP BY and HAVING clauses, ORDER BY clause.
9.	Implementation of Procedure.(Basic procedure, procedure for Insert, Update and Delete)
10	Implementation of Triggers(validation before and after insert, delete and update)